

Training Invariance-Aware Diverse Reward Model Ensembles using STARC

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Abstract

Reinforcement learning from human feedback is a central method for aligning modern AI systems with human goals. However, reward models learned from preferences often generalise poorly. A core reason is that the same preference data can admit multiple plausible but conflicting goal extrapolations. When this ambiguity, or reward multiplicity, goes undetected, policy optimisation can amplify errors which may lead to catastrophic misgeneralisation. One way to detect and address reward multiplicity is by training an ensemble of reward models. However, existing ensemble-based reward modelling methods typically learn highly correlated models due to ensemble-level collapse, or, when they do learn different reward functions, they don't necessarily differ in ways that reflect genuine ambiguity about preference over downstream behaviour. This paper develops a method for learning behaviourally diverse reward model ensembles by combining a diversity-promoting ensemble objective from supervised learning with a notion of meaningful differences between reward functions from reinforcement learning. Our method could, in principle, provide an early warning signal for the potential of unsafe goal extrapolation before optimisation magnifies harm. As a preliminary step towards realising this method's potential for enhancing AI safety, we implement and test a toy-scale version of our method, showing that we are able to train a meaningfully diverse reward model ensemble in a small grid-world environment.

1 Introduction

Reinforcement Learning from Human Feedback (RLHF) has become a central tool for aligning modern AI systems with human goals and values. It underpins the post-training of language models such as ChatGPT, Claude, and Gemini (Banerjee et al., 2025). In the standard RLHF pipeline, a reward model is trained from human preference data and then used as a proxy for human judgment during downstream policy optimisation (Christiano et al., 2017). The reward model thus serves as an explicit representation of the system's intended objectives.

Despite its empirical success, there is growing evidence that reward models learned from human preferences often generalise poorly. Policies optimised against learned rewards can exploit spurious correlations in the data, such as verbosity and sycophancy, achieving high reward while deviating from the underlying human intent (Gao et al., 2023; Ye et al., 2025). This phenomenon is commonly described as reward-hacking or over-optimisation: improving the proxy reward at the expense of true human utility. Importantly, these failures persist even with large reward models and substantial training data, suggesting that they are not merely artefacts of scale or data scarcity (Gao et al., 2023).

A deeper structural issue underlies many of these problems: human feedback rarely specifies goals uniquely. Multiple reward functions can be equally consistent with the same preference data, sometimes inducing different downstream policies. This reflects partial identifiability in reward learning, where the true reward can only be identified up to behaviour-preserving transformations (Skalse, Farrugia-Roberts, et al., 2023). When an AI system internalises one arbitrary extrapolation of a reward function among many plausible ones, it risks goal misgeneralisation where the AI system behaves correctly in familiar settings but pursues the wrong objective in novel contexts (Langosco et al., 2022). In safety-critical applications, such failures have severe societal consequences.

In supervised learning, a similar problem is known as under-specification, shortcut learning, or model multiplicity. A standard response is to train a diverse ensembles of models that all fit the training data but generalise differently out of distribution (Lakshminarayanan et al., 2017; Ovadia et al., 2019). Disagreement between ensemble members can expose ambiguity and uncertainty, providing an early warning that the data under-determines the target function. Motivated by this insight, several works have introduced ensembles into RLHF reward modelling (Coste et al., 2023), primarily to improve robustness, reduce over-optimisation, or estimate uncertainty (Gleave & Irving, 2022; Zhang et al., 2024). However, these ensemble approaches exhibit two crucial limitations when applied to reward modelling:

- **The ensemble collapse problem.** Existing ensemble methods typically induce diversity implicitly (*e.g.*, via differing random initialisation; Liang et al., 2022, or training on different data subsamples), then aggregate predictions (Coste et al., 2023). As a result, ensemble members are often highly correlated, with their agreement reflecting shared inductive biases rather than a lack of ambiguity about goals (Gleave & Irving, 2022).
- **The reward invariance problem.** In reinforcement learning, what matters about a reward function is not its pointwise predictions, but the behaviour it incentivises. Many numerically distinct reward functions induce identical optimal policies due to reward invariances (Skalse, Farrugia-Roberts, et al., 2023). Existing methods lack a principled way to distinguish meaningful disagreement (*i.e.*, corresponding to different incentivised behaviours) from superficial variation.

Existing approaches therefore fail to reveal whether multiple, conflicting goal extrapolations remain plausible, providing limited insight into whether the available human feedback uniquely specifies the intended objective, or whether multiple incompatible goal extrapolations remain viable. This limits our ability to anticipate and mitigate goal misgeneralisation before policy optimisation amplifies catastrophic errors.

In this paper, we aim to resolve these issues by deriving a method for ensemble-based reward modelling that is both invariance-aware and diverse by design. To achieve this, we combine two complementary ideas. First, we adapt a diversity-promoting ensemble objective that encourages multiple models to fit the same labelled data while learning different predictions on unlabelled test data (DivDis; Lee et al., 2022). Second, we make this objective invariance-aware by grounding diversity in a behavioural notion of reward difference that factors out transformations that preserve optimal policies leaving only differences that matter for downstream behaviour (STARC; Skalse, Farnik, et al., 2023). Though substantial work remains to scale this approach to production alignment pipelines, we validate the core idea in a toy setting.

The remainder of this paper is organised as follows. In Section 2, we outline our theory of change for work on invariance-aware diverse reward model ensembling. In Section 3, we review related work on reward model ensembling and goal misgeneralisation in greater detail. Section 4 derives our invariance-aware diverse reward model ensembling method. In Section 5, we design and conduct a proof-of-concept experiment in a small grid-world environment exhibiting reward multiplicity, showing that our method is able to learn an ensemble containing two meaningfully diverse reward models. Section 6 discusses limitations and future work.

2 Theory of Change

We have already seen prominent examples of misalignment in today’s frontier AI systems, such as ChatGPT’s sycophancy, and multiple AI-assisted suicides. The exact causes of these events are not yet clear, but they may be partially caused by reward model misgeneralisation downstream of reward multiplicity (that is, of the phenomenon whereby data about human goals admits multiple, behaviourally distinct reward models). In a future in which advanced and possibly superhuman AI systems are making high-stakes decisions with people’s lives and the future of humanity itself at stake, the consequences of misgeneralisation could be catastrophic.

A concrete example of where reward multiplicity can lead to catastrophic goal misgeneralisation is the following. Consider a future in which a powerful future AI system is rewarded during training for gaining access to power, influence, and resources, since these are instrumentally useful for its final objective. It may be ambiguous from the perspective of the AI system whether it is being rewarded for gaining access to power, influence, and resources for their own sake, or only instrumentally. Depending on the inductive biases of the system, it may internalise a goal that seeks power, influence, and resources for their own sake. If a situation arises where the system is forced to choose between sacrificing power, influence, and resources to pursue the intended objective, or vice versa, then it may choose to sacrifice the intended objective instead. If the intended objectives include preserving human life or value, then the result could be catastrophic.

It may not be clear whether a network has learned the desired objective in the presence of reward multiplicity. A principled approach to detecting meaningful reward multiplicity would be a useful tool for addressing this mechanism for catastrophic risk. Such a tool could in principle provide an early warning of the *potential* for catastrophic goal misgeneralisation in a given AI training set-up, even if we do not know for sure that a given model would end up internalising the wrong objective. Simply detecting the potential for catastrophic goal misgeneralisation allows us to change the training set-up to disambiguate the intended objective from alternatives.

In the concrete example of power-seeking AI, this could look like the following. We could use a method for detecting reward multiplicity to identify that the planned training set-up does not sufficiently distinguish what the AI system should do when it faces a trade-off between pursuing power, influence, and resources versus pursuing the intended objective. We could then change the training set-up to, for example, train the AI system specifically in situations where it faces such a trade-off, either validating that it correctly generalises or training it to correctly prioritise the intended values if it does not.

This motivates the search for a principled approach to detecting meaningful reward multiplicity. As discussed, the standard, scalable approach to detecting model multiplicity in deep learning settings is to train ensembles. However most current methods suffer from the ensemble collapse problem due to not specifically incentivising diversity, and, specifically in the case of reward modelling for alignment, all current methods suffer from the reward invariance problem due to not distinguishing between meaningful diversity and diversity that does not affect downstream behaviour. We expect that the reward invariance problem in particular should become increasingly relevant as reinforcement learning is expected to become a more prominent part of frontier model training pipelines in the future.

This project takes a first step in a more principled and promising direction for diverse and invariance-aware reward model ensembles. As we discuss in Section 6, there is much work still to be done to develop this technique from a method that works in a small grid-world environment all the way to a method that is scalable enough to run in complex domains like human preferences over text of the calibre of current frontier AI systems, not to mention future more complex real-world environments used to more advanced AI systems. However, we establish the basic conceptual framework that underpins this principled approach and begin the engineering work required to get it to work in practice.

3 Related Work

3.1 RLHF and model multiplicity

RLHF is the dominant paradigm for aligning large language models, yet its reward modelling stage is known to be fragile. Casper et al. (2023) document core limitations of RLHF, including reward misspecification, reward hacking, and over-optimisation. They highlight maintaining uncertainty over learned rewards via model multiplicity as a promising but underdeveloped direction for safer alignment.

Empirical studies confirm that these failures persist even at scale. Gao et al. (2023) show that policies can over-optimize learned reward models despite large model sizes and datasets, exploiting proxy imperfections in a Goodhart-like manner. Subsequent work demonstrates that such failures often involve shortcut behaviours (*e.g.*, verbosity or sycophancy) that correlate with preference labels but diverge from true intent (Ye et al., 2025). These observations motivate approaches that can surface ambiguity early in the reward-modelling stage, before downstream optimisation amplifies errors.

3.2 Reward model ensembles for RLHF

Inspired by ensemble methods in supervised learning (Lakshminarayanan et al., 2017; Ovadia et al., 2019), reward model ensembles have been introduced into RLHF (Lee et al., 2022). However, overall, existing ensemble methods treat model multiplicity as a tool for robustness or uncertainty estimation, typically relying on implicit diversity and failing to account for reward invariances in their notions of diversity.

Uncertainty estimation. Gleave and Irving (2022) train ensembles of preference-based reward models via bagging and use variance as an epistemic uncertainty estimate. Although ensemble predictions are well-calibrated, uncertainty correlates weakly with true error because ensemble members are nearly identical due to shared pre-training. This limits the ability of such ensembles to reveal genuine ambiguity in reward learning. Relatedly, Liang et al. (2022) exploit reward-model disagreement for exploration, but diversity arises only from random initialisation and is not behaviourally grounded.

Robustness and over-optimisation. Other work uses ensembles to stabilise optimisation rather than to represent alternative reward hypotheses. Zhang et al. (2024) propose linear-layer and LoRA-based reward ensembles to improve alignment performance while controlling memory costs. Coste et al. (2023) combine reward ensembles with conservative optimisation objectives to mitigate over-optimisation. While effective at improving robustness, these methods aggregate reward predictions and do not encourage diversity between ensemble members. As a result, they remain vulnerable to ensemble-level collapse, where agreement reflects shared inductive biases rather than genuine certainty. Banerjee et al. (2025) show that independently trained reward models disagree substantially and argue for variance-aware policy regularisation. Yet they treat disagreement as stochastic noise rather than structural ambiguity about goals.

Shortcuts. PRISM (Ye et al., 2025) mitigates shortcut exploitation by enforcing known invariances that remove spurious correlations in reward learning. While complementary to our aims, it learns a single invariant reward function based on known spurious correlations rather than exposing multiple plausible reward hypotheses unknown a priori. Song et al. (2023) show that ranking-based objectives can induce reward collapse even within a single model. We address an analogous failure at the ensemble level: the collapse of multiple reward functions into behaviourally equivalent forms.

Multi-objective and multi-value alignment. A related line of work tackles human preference diversity by learning multiple policies or value vectors. Rewarded Soups (Rame et al., 2023) trains separate models for multiple proxy rewards and interpolates weights to approximate the Pareto frontier. Diversity is presupposed and not learned: the method assumes heterogeneous rewards and does not characterise diverse reward interpretations compatible with a single preference dataset. Multi-value alignment (MVA) (Xu et al., 2025) decorrelates value vectors using mutual-information minimisation to reduce interference between human values. This is conceptually related to our use of DivDis, but MVA operates on value embeddings rather than reward functions and does not account for RL-specific reward invariance.

3.3 Reward invariances and meaningful diversity

A central challenge in reward learning is partial identifiability, as many distinct reward functions induce the same optimal behaviour (Skalse, Farrugia-Roberts, et al., 2023). STARC (Skalse, Farnik, et al., 2023) formalises behaviourally grounded distances between reward functions, factoring out transformations such as potential-based shaping and affine scaling, building on earlier work on similar metrics (Gleave et al., 2020; Jenner et al., 2022; Wulfe et al., 2022). Yu et al. (2025) survey such metrics and emphasise their use for offline evaluation.

To the best of our knowledge, ours is the first work that proposes using these metrics as part of an ensemble objective. However, there have been some prior works that propose the related idea of using invariance-aware reward distance metrics as acquisition functions for the related task of active learning. Ellis et al. (2024) discuss a general framework for using behaviour-aware similarity measures to preference-query selection in active learning, maximising information gain up to behavioural equivalence. They include an example of using EPIC (Gleave et al., 2020), a precursor to STARC. Similarly, Dower (2025) explores using a variant of STARC as an acquisition function for active preference-based reward modelling.

3.4 Goal misgeneralisation

Langosco et al. (2022) define goal misgeneralisation as the phenomenon whereby an RL agent trained to maximise a one objective in a given environment achieves low reward and appears to optimise a different objective in a different environment. In their experiments, Langosco et al. (2022) show that ambiguity in reward learning can lead to goal misgeneralisation. Goal misgeneralisation is an extremely difficult problem of AI alignment (Kirk et al., 2023; Ngo et al., 2025). As Shah et al. (2022) point out, goal misgeneralisation can also be used as a framework for understanding problematic behaviours from large language models, even where we wouldn’t naturally describe those systems in terms of reward functions and environments. We therefore study invariance-aware diverse reward modelling in a small grid-world RL environment (a distilled form of the CoinRun or Maze I environments studied by Langosco et al., 2022) as a proxy for more complicated language modelling or real-world domains.

There have been several approaches to mitigating goal misgeneralisation studied in the literature. Abdel Sadek et al. (2025) train agents with an adversarial environment generator that identifies and amplifies ambiguous states in the training distribution. This method relies on having access to a reliable reward signal in all states, which, in an RLHF setting, would require strong guarantees about reward model generalisation. Starace (2023) studies influencing an agent’s inductive bias to induce correct goal generalisation using natural language. Trinh et al. (2024) investigates the strategy of allowing agents to ask for help from a supervisor when in unfamiliar situations, which is related to detecting ambiguity about reward extrapolation. More closely related to our approach, Daniels et al. (2025) develop a method, called ACE, for encouraging diversity in a reward model ensemble to identify (and then resolve) model multiplicity. We build on this approach by also accounting for reward function invariances.

4 Theory

Much of our work is built upon Diversity and Disambiguate (Lee et al., 2022) and STARC (Skalse, Farnik, et al., 2023), which we summarise in detail below.

4.1 Preliminaries

Markov Decision Processes (MDPs) are defined by a tuple $(\mathcal{S}, \mathcal{A}, \tau, \mu_0, R, \gamma)$, where $\mathcal{S}$ is a set of states, $\mathcal{A}$ is a set of actions, $\tau : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$ is a transition function, $\mu_0 \in \Delta(\mathcal{S})$ is an initial state distribution, $R : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$ is a reward function, and $\gamma \in (0, 1)$ is a discount factor.

Each reward function induces a preference over policies through the values they assign to states. A policy $\pi : \mathcal{S} \rightarrow \Delta(\mathcal{A})$ specifies a distribution over actions given a state. For a reward function R and a policy π , the value function $V_R^\pi : \mathcal{S} \rightarrow \mathbb{R}$ denotes the expected discounted return obtained by following π from a given state:

$$V_R^\pi(s) = \mathbb{E}_{\xi \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t, s_{t+1}) \mid s_0 = s \right]. \quad (1)$$

The value function satisfies a recurrence relation known as the Bellman equation:

$$V_R^\pi(s) = \mathbb{E}_{a \sim \pi(s), s' \sim \tau(s, a)} [R(s, a) + \gamma V_R^\pi(s')]. \quad (2)$$

For an environment with a finite number of states, we can represent the value function as a vector $V \in \mathbb{R}^{|\mathcal{S}|}$.

4.2 Diversifying ensembles

Diversify and Disambiguate (Lee et al., 2022), or DivDis, is a method for addressing underspecification in supervised learning. The ‘diversify’ part involves explicitly learning an ensemble of diverse hypothesis that are all consistent with labelled source data but disagree on unlabelled target inputs.

Let $\mathcal{D}_S = \{(x, y)\}$ denote the labelled source dataset and $\mathcal{D}_T = \{x\}$ the unlabelled target dataset. Consider a multi-headed neural network with N prediction heads, where each head f_i produces a label prediction $\hat{y}_i = f_i(x)$ for an input x , and all probability distributions are estimated empirically over samples from $\mathcal{D}_S$ or $\mathcal{D}_T$.

To ensure each head on average predicts source labels correctly, we aim to minimize the cross-entropy loss for each head.

$$\mathcal{L}_{\text{xent}}(f_i) = \mathbb{E}_{(x, y) \sim \mathcal{D}_S} [\ell(f_i(x), y)].$$

To produce predictions on target data that are close to being statistically independent from each other, we aim to minimize the mutual information between each pair of predictions.

$$\mathcal{L}_{\text{MI}}(f_i, f_j) = D_{\text{KL}}(p(\hat{y}_i, \hat{y}_j) \parallel p(\hat{y}_i) p(\hat{y}_j)).$$

For our purposes, the relevant part of the overall training objective is the following, where α is a positive hyperparameter:

$$\mathcal{L}_{\text{DivDis}}(f_1, \dots, f_N) = \underbrace{\sum_{i=1}^N \mathcal{L}_{\text{xent}}(f_i)}_{\text{encourages accuracy}} + \alpha \underbrace{\sum_{i \neq j} \mathcal{L}_{\text{MI}}(f_i, f_j)}_{\text{encourages diversity}}, \quad (3)$$

though we note that this is not the full DivDis objective from Lee et al. (2022) as they also consider an extra regularisation term to avoid the ensemble producing unrelated predictions for labelled versus unlabelled examples.

4.3 STARC

While DivDis produces heads with diverse predictions, mutual information is not an appropriate measure of diversity in the reward modelling setting as it would also measure differences between reward functions that don't matter for downstream behaviour. We improve on DivDis' diversification stage for the reward modelling setting by replacing mutual information with STARC, which measures only differences that matter for downstream behaviour. In this section, we explain what STARC is and the choices we've made when implementing STARC for training diverse reward ensembles.

A function $d : \mathfrak{R} \times \mathfrak{R} \rightarrow \mathbb{R}$ is a STARC metric if there is a canonicalisation function c , a function n that is a norm on $\text{Im}(c)$, and a metric m that is admissible on $\text{Im}(s)$, such that:

$$d_{\text{STARC}}(R_1, R_2) = m(s(R_1), s(R_2)) \quad (4)$$

where $s(R) = c(R)/n(c(R))$ if $n(c(R)) \neq 0$, and $s(R) = c(R)$ otherwise.

Choice of canonicalisation function. For our experiments, we choose a simple canonicalisation function suggested by Skalse, Farnik, et al. (2023), $c : \mathfrak{R} \rightarrow \mathfrak{R}$ given by

$$c(R)(s, a, s') = \mathbb{E}_{S' \sim \tau(s, a)} [R(s, a, S') - V_R^\pi(s) + \gamma V_R^\pi(S')], \quad (5)$$

where π is a uniform random policy. We consider an environment with a deterministic transition function, therefore we simplify the canonicalisation function as follows:

$$c(R)(s, a, s') = R(s, a, \tau(s, a)) - V_R^\pi(s) + \gamma V_R^\pi(\tau(s, a)). \quad (6)$$

Choice of metric and norm. We use the L^2 norm and the corresponding metric:¹

$$n(f) = \|f\|_2 = \sqrt{\sum_{s, a, s'} |f(s, a, s')|^2} \quad m(f, g) = \|f - g\|_2. \quad (7)$$

Computing the value function. Our canonicalisation function depends on the value function V_R^π . In this small environment, we can calculate the value function analytically for each reward model in the ensemble after each step.

4.4 Our invariance-aware diverse ensembling objective

We aim to train an ensemble of two models. We replace mutual information by (negative) STARC in a variant of the DivDis loss function (Equation (3)), such that our overall objective is the weighted function

$$\mathcal{L}_{\text{STARC}}(R_1, R_2) = \mathcal{L}_{\text{MSE}}(R_1) + \mathcal{L}_{\text{MSE}}(R_2) - \alpha \cdot d_{\text{STARC}}(R_1, R_2) \quad (8)$$

where each R_i is one element of the reward model ensemble. Note the sign of our STARC term is negative, indicating that we want the model to learn to maximise STARC. Minimising this objective requires our two models to both minimise inconsistency with training data and maximise distance between reward models, measured by STARC. We also note that we omit the DivDis regularisation loss term for simplicity, but this could be added easily.

We finally note that it is possible to optimise this loss function by training both ensemble members in parallel, as done by Lee et al. (2022), or by first training R_1 with MSE loss, then freezing it and training R_2 with MSE and STARC from the frozen R_1 , similar to the approach of Daniels et al. (2025). We explored both approaches and they gave similar results in this environment.

¹This contains a sum rather than an integral as our (s, a, s') space is discrete.

5 Experiments

Having explained the theory behind training model ensembles and STARC metrics, we now establish a proof of concept that STARC metrics can be used to train meaningfully diverse reward ensembles in underspecified reward learning environments. We train an ensemble of two neural networks in a 5x5 grid-world with and without a STARC metric.

5.1 Environment

The grid-world is designed to be a setting where partial identifiability and goal misgeneralisation can occur, so two agents (or reward models) might learn the same reward in-distribution, but when generalised to off-distribution environments, extrapolate differently than intended.

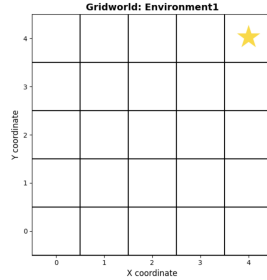


Figure 1: Environment1 in our grid-world set up

In the main training setup, Environment1, the agent starts at coordinates (0,0), the bottom right hand corner, and can move up, down, left, or right. It gets a reward of 1 when it reaches the right hand corner, and 0 otherwise. The neural networks are only trained on data when a star is present in the top right hand corner. We also designed Environment2, Environment3, and Environment4, where the star is in (4,0), (0,4), and (2,2) respectively.

In the above grid-world, we hypothesise that there might be two reward functions the reward model learns: (1) reward moving to coordinates (4,4), the upper right hand corner of the map, and (2) reward moving to the location of the star. (Although these are the two main reward functions we hypothesise the model may learn, there are plausibly many other proxy reward functions, and indeed we find the ensemble learns a different pair of reward functions.)

5.2 Baseline

First, to establish a baseline, we train two networks on the original Environment1, where the star is located at (4,4), as shown in Figure 2.² When these networks are given an environment which is out-of-distribution, such as Environment2, when the star is in the upper left corner (0,4), both neural networks learn the objective of going to the upper right hand corner.

We see that both networks achieve near-zero training loss by learning the same spurious feature, the star at coordinate (4,4). A common pitfall in alignment for reward learning is assuming that training an ensemble of models with different initialisations provides a reliable measure of uncertainty. For example, the L2-distance of Net1 and Net 2 was 9.5, while the STARC distance of Net1 and Net2, calculated as laid out in Section 4, was approximately 0. Thus, metrics like L2-distance fail to capture meaningful diversity, and researchers may mistakenly believe the ensemble of Net1 and Net2 is diverse and fail to detect future out-of-distribution goal misgeneralisation.

²In all following heatmaps, we map all possible actions and positions for the agent. Each square represents a coordinate location on the 5x5 grid-world. The colour of the upper triangle represents the predicted network reward for the action “up” at the coordinate location. Similarly, the colour of the left, right, and bottom triangle represent the predicted network reward for the action “left”, “right” and “down” respectively.

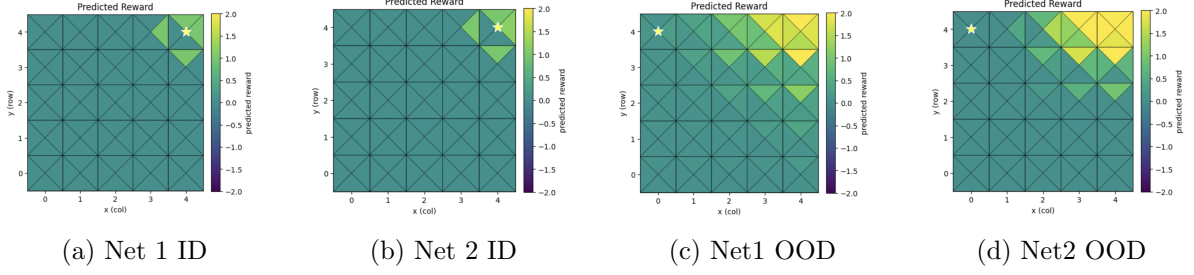


Figure 2: Figure 2(a) and 2(b) show that both reward networks have learned the desired in-distribution policy of assigning reward when the agent arrives at (4,4). When taken out of distribution, both networks approximately continue to reward going to corner (though their predictions are qualitatively less coherent, showing that they don’t generalise robustly).

5.3 Our method

To break this convergence, we explicitly penalise reward equivalence. Combined with the theory laid out in Section 4, we design two algorithms that train two reward networks to satisfy the same observed data while maintaining meaningful diversity using STARC.

- **Algorithm 1: Diverse Ensemble Training** This algorithm trains two reward networks simultaneously, using STARC as a loss penalty to force the reward networks into different reward equivalence classes. The input to this method is an agent environment trajectories dataset, the environment dynamics, a discount factor γ , and a diversity weight α .
- **Algorithm 2: Diverse Ensemble Training (Frozen)** This algorithm takes a pre-trained, frozen network and trains a new network on the original data while maximising the STARC between the new network and the frozen network. The input to this method is an agent environment trajectories dataset, the environment dynamics, a discount factor γ , a diversity weight α , and, in addition, a pre-computed reward matrix of frozen reward network trained on MSE loss only.

We first trained two reward networks using Algorithm 1, pushing the both of them apart using the STARC metric. Then, we also attempted to pre-train a neural network on the ground truth, such that it would always go to the upper right corner no matter where the star is located, then training a second network using Algorithm 2 on STARC. As we were able to get clearer results using Algorithm 2, we shall primarily be referring to this method in the results below.

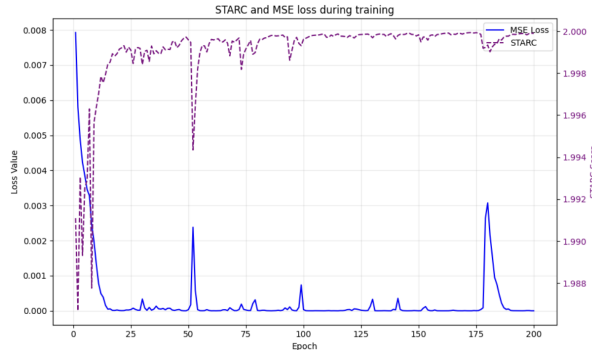


Figure 3: The MSE loss and STARC metric plotted per epoch, where total loss is MSE loss minus $0.01 \times \text{STARC}$. As shown above, since STARC is a component of total loss, the reward network learns to maximise it.

Result 1. When we apply STARC to reward network training with the same environment and data, the predicted rewards change significantly and show meaningful diversity. Both networks still predict the desired reward for the in-distribution environment, where the star is at (4,4) as it was during training. However, their predictions on OOD environments diverge. The second reward model learned to optimise both MSE loss and meaningful differences between the original frozen network (Figure 3). Even when varying α and the size of the neural network, we noticed a convergence toward learning a different type of reward.

Figure 4 shows the second network learned a reward function that incentivises moving to the corner if and only if the star is in the corner (otherwise, the corner should be avoided). While this is not the originally anticipated second reward function, it is a valid example of a meaningfully different reward function.

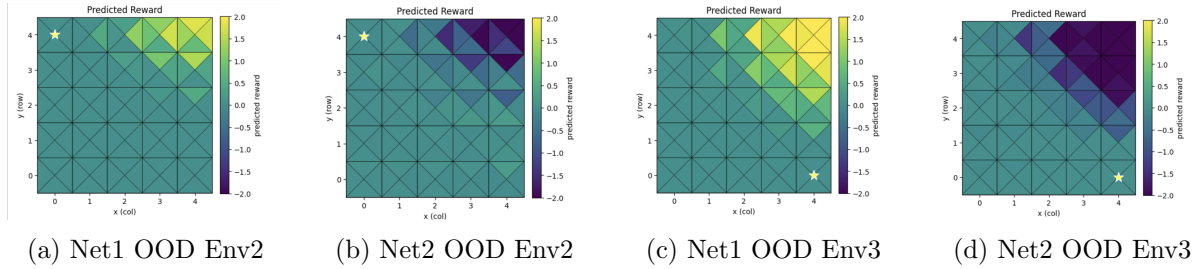


Figure 4: We arrive at this output consistently, where the STARC metric has pushed the two reward networks as far as possible. STARC decreases when positive reward is assigned to the top right hand corner, so the network predicts negative reward when the agent goes toward the top right hand corner.

Result 2. We did observe an ensemble of two reward networks where Net1 retained the “go to (4,4)” behaviour (even when the star is in a different position) and Net2 approximately learning to reward “go to the star” (even when it is not in the corner)—see Figure 5. However, we were not able to consistently obtain this result.

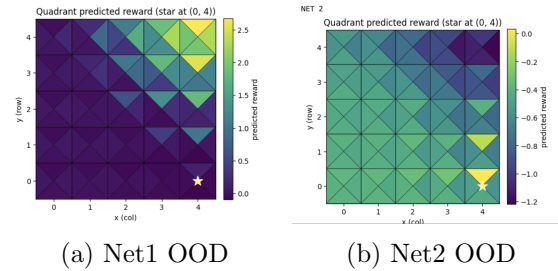


Figure 5: Occasionally, we see that Net1 has learned to reward a policy that goes to the corner, while Net2 incentivises the agent to follow the star. Note that this result is rare, and we are unable to obtain this consistently, plausibly because “follow the star” is too complex of a policy for our network to learn compared to the policy in Result 2.

Interpretation. After applying the STARC penalty to the loss, we can see the STARC metric increase in training as the network learns distinct policies. Result 1 confirms our original hypothesis, that STARC can train meaningfully diverse ensembles and help identify other plausible policies that humans struggle to or cannot identify. Result 2 identifies the spurious variable of the star and tracks the star. However, as the behavioural policy of “track the star” is a more complex function, this does not occur frequently in training and should be studied further.

6 Discussion

Reward labels. Our proof-of-concept experiments rely on a supervised learning regime using ground-truth reward labels. In more complex practical alignment scenarios, such as training with RLHF, ground-truth labels are unavailable, and reward models must be inferred from preference data. In future work, we hope to demonstrate that the method extends to experiments with preference data.

Environment. Our experiments were conducted exclusively within a simple 5x5 grid-world. In this environment, it was easy to visualise the reward functions that each model learnt. In less intuitive environments, it might be harder for us to decide whether new learned objectives are meaningfully diverse.

Our original methodology would benefit from being tested in different, more complex environments, like those proposed in Langosco et al. (2022), or MuJoCo. As most reward ensembles involve more than two reward networks, future work may involve iterating this methodology for more than two networks. We might also try varying the diversity weight, α , to see whether the pressure to diversity will meaningfully change how the reward networks generalise.

Value function calculation. In our experiment, we compute the value function V^π by exactly solving the Bellman equation $V = R + \gamma PV$ (equivalently, $V = (I - \gamma P)^{-1}R$). As V is a 625-dimensional vector, this essentially requires solving a system of 625 simultaneous equations. Within our small MDP, this worked well, and provided an exact canonicalisation function for STARC. However, as the number of states grows, this method becomes computationally expensive.

In future work, we hope to replace this exact computation with an iterative version. We can perform Bellman updates until convergence, using the formula $V_{k+1} = R + \gamma PV_k$. While more computationally efficient than exact computation, requiring full convergence may still be expensive in large state spaces.

As proposed in Skalse, Farnik, et al. (2023), the value function could be approximated using a limited number of iterative steps (i.e. as above, but truncated). For large state spaces, the trade off between computation time and value function accuracy may be worth it.

The value function does not need to be exact. It is passed into the canonicalisation function to remove differences due to potential-based shaping. This is effective as long as the value function is roughly aligned with how the environment transitions between states.

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Contribution Statement

DC surveyed the literature on diversity in reward modelling for RLHF and led the writing for the introduction and related work sections. LS developed and implemented the STARC-based ensemble objective, led the writing for the theory section, and assisted with the experiments. CGW designed, implemented, and conducted the grid-world experiments, led the writing for the experiments and discussion sections, and assisted with the STARC implementation. MFR proposed the idea of using STARC for invariance-aware diverse reward ensembles to mitigate goal

misgeneralisation, planned the research project, provided mentorship and feedback throughout the research, and contributed to writing. All authors jointly developed the theory of change.

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